

Ques. WHY IS BATTERY CHARGING FROM THE USB PORT OF A PC OR LAPTOP SLOW?

Suresh D.

Ans. This is one of the common questions asked by most mobile users. Depending on the usage and battery capacity of your mobile phone, you need to charge its battery at least once a day or sometimes more often. Most mobile charger cables have USB type-A connector on one end. You just need a wall adaptor with USB port or computer USB port to charge the mobile battery. But there is a differ-

ence in charging speed between wall adaptor and computer. chargers are rated at 5V and 1000mA (1A), so their power output would be $5 \times 1 = 5$ watts. Some wall chargers have even higher current rating. Thus battery charging from the USB port of a PC or laptop is comparatively slow.

Q2. HOW DO MOBILE PHONES PRODUCE ELECTROMAGNETIC RADIATION?

Kone Prasanna Narayan

A2. Electromagnetic radiation refers to the waves of electromagnetic field radiating through space. This radiation includes radio waves, microwaves, ultraviolet, infrared, X-rays and gamma rays.

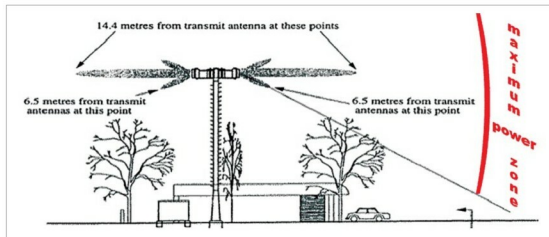
microphone and a radio signal is generated from the mobile's circuitry. The radio signal is transmitted through its antenna, which radiates through space. This transmitted radio wave is picked up by the antenna at the base station. Then the mobile phone and base station start communicating back and forth through radio waves transmitted from their respective antennae.

Every antenna, either on a mobile phone or tower, radiates electromagnetic waves. Radio waves are the strongest near the mobile phone antenna but get weaker and weaker as you move away from the phone. Similarly, radio waves are the strongest at the base station antenna.

The radiated power level near a tower is high but reduces as you move away from it.

Typically, a base station may have several antennae mounted on a tower. The tower or mast itself does not radiate any energy; it is radiated from antennae mounted on it.

The maximum power near the tower is shown in



A base station (Courtesy: <http://www.steeletheair.com.au>)

ence in charging speed between wall adaptor and computer.

The charging speed depends on various factors, including the power rating of the charging source. The power can be determined by multiplying the voltage and the current supplied by the charger.

Normally, the USB port of your PC gives 5V and up to 500mA (0.5A), which means delivers about 2.5-watt power. But if your PC has USB3.0 port, normally rated at 900mA current, you can get higher power. Some wall

A mobile phone has both transmitter and receiver sections. When the mobile phone is turned on, it emits radio waves that consist of radio frequency (RF) energy—a form of electromagnetic radiation moving at the speed of light. It works by transmitting radio wave signals to (and receiving these from) nearby base stations. (Base station is an essential part because it links the individual mobile phone with the rest of the network.)

When talking on a mobile phone, your voice is captured through its

the figure on this page. The tower should be tall enough, as shown in the figure, to avoid high power radiation to nearby areas. The cellular or mobile network in a city may have a large number of base stations. The more the number of antennae, the more is the radiation power intensity in the nearby area.

Answers compiled by Sani Theio, technical editor. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender.